



Aligning digital strategy and resources to create Healthcare 4.0 organizations: Implications for smart operations and smart working

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ABSTRACT

This study investigates the organizational foundations required to implement Healthcare 4.0 (H4.0) by examining how digital strategy, internal and external digital resources, and network embeddedness support the development of smart operations and smart working in healthcare organizations. While prior research has predominantly focused on technology-driven improvements in healthcare processes, less attention has been given to the sociotechnical integration of people, processes, and technologies, particularly from a strategic and managerial perspective. Drawing on a survey of 303 hospitals, we assessed the role of digital strategy and digital resources in advancing H4.0, conceptualized through the dual lenses of smart operations (technology for process automation) and smart working (technology to support workers). Using hierarchical regression and bootstrapping techniques, we find that digital strategy positively influences both smart operations and smart working. However, internal and external digital resources mediate this relationship only in the context of smart working. This indicates that while process automation relies primarily on strategic clarity, supporting workers requires stronger resource structures. Additionally, we find that healthcare network embeddedness positively moderates the relationship between digital strategy and external digital resources, but not internal resources. This suggests that external collaborations enhance a hospital's ability to translate strategic intent into actionable use of external resources, particularly in support of smart working. Our findings contribute to a deeper understanding of how strategic alignment and network embeddedness shape sociotechnical systems in healthcare. They offer guidance for managers seeking to design holistic digital transformation strategies to build H4.0 organizations.

1. Introduction

The digital transformation of healthcare organizations has been characterized by the application of advanced digital technologies such as the Internet of Things, Cloud Computing, Big Data, and Artificial Intelligence in medical devices and healthcare information systems (Kraus et al., 2021), resulting in what is called as 'Healthcare 4.0' (H4.0) (Thuemmler and Bai, 2017). The introduction of these technologies enables real-time personalization of patient and workers care services, facilitating the transition to a patient-centered environment (Wang et al., 2018). This transformation involves the interaction of people, processes, and digital technologies, forming micro-foundations of operational capability (Roscoe et al., 2019). In this sense, the concept of H4.0 considers the interaction of two main dimensions that form a

sociotechnical healthcare system: from the technical side, the use of digital technologies to automate operational processes is commonly referred to as *healthcare smart operations* (Tortorella et al., 2020a), while the social side considers the use of digital technologies to support healthcare worker activities, usually known as *healthcare smart working* (Klein and Watson-Manheim, 2021). When H4.0 is conceived to create a balanced sociotechnical system between workers, organizational processes, and technologies, both smart working and smart operations need to be developed as part of the healthcare digital transformation strategy (Grover et al., 2018).

The literature has focused on understanding which digital technologies are used and how digital transformation can contribute to achieving better performance outcomes in healthcare systems (Tortorella et al., 2022a; Tortorella et al., 2022b). Overall, this literature

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has been predominantly oriented toward the techno-centric view of smart operations. For example, [Dicuonzo et al. \(2023\)](#) explore how artificial intelligence can support the process of the healthcare system; [Cannavacciuolo et al. \(2023\)](#) address the organizational changes through the implementation of telemedicine in healthcare; and [Basile et al. \(2024\)](#) consider the role of big data analytics to improve the quality of healthcare services. These studies are concerned with how digital technologies can boost different operational processes. On the other hand, an emerging stream has highlighted the need for integrating people with processes and technologies, considering that there is a sociotechnical interplay between them ([Grover et al., 2018](#)). For instance, [Klinker et al. \(2020\)](#) investigate the use of augmented reality technologies to increase healthcare workers' capabilities and [Naslund et al. \(2019\)](#) consider the use of digital technologies to scale up healthcare through nonspecialist workers. These are a few of the cases when digital technologies are considered for smart working in healthcare, bringing a social perspective to the technological view. However, the literature has missed the consideration of what is necessary in the context of strategy and organizational design when smart operations and smart working are in the spotlight of Healthcare 4.0.

When discussing the adoption and use of digital technologies, the influence of strategic and managerial aspects is often overlooked ([Chanias et al., 2019](#); [Verhoef et al., 2021](#)), particularly in terms of their roles to balance the organizational system from a sociotechnical perspective. In this context, achieving organizational clarity is essential to support the effective adoption of digital technologies ([Marcon et al., 2022](#); [Marcon et al., 2025](#)). Otherwise, the adoption of these technologies may be unsuccessful or limited to isolated implementations ([Zimmer et al., 2023](#)). This organizational support has often been regarded as a matter of establishing a digital strategy and managing digital resources (e.g., technological infrastructure, people skills, and financial investments for digitalization) within organizations ([Verhoef et al., 2021](#); [Franco et al., 2024](#); [Correia and Frank, 2025](#)). In this context, the vision of strategic alignment between strategy and resources is essential ([Edelman et al., 2005](#)). This is because H4.0 can be supported through an appropriate digital strategy, backed by suitable resources, and, as a result, can facilitate the adoption of healthcare smart operations and healthcare smart working concepts. The alignment between strategy and resources can be further strengthened when an organization is embedded within a network of external healthcare actors ([Correia and Frank, 2025](#)). Such embeddedness facilitates access to critical knowledge and resources through ongoing interactions ([Granovetter, 1985](#)), including those that support digitalization efforts ([Xing et al., 2023](#)). In this context, network embeddedness represents a key component of the environmental dimension of sociotechnical systems that underpin digital transformation processes ([Marcon et al., 2024](#)).

However, little is known about the way digital strategy, resources and the connection with the external environment are associated in the implementation of healthcare sociotechnical systems in the context of digital transformation. It is well known that organizations need to define strategies for digital transformation ([Chanias et al., 2019](#)) and that managing resources is necessary for the effective implementation of digital solutions ([Dalenogare et al., 2023](#)), but it is not clear the role these organizational elements play in the development of smart operations and smart working concepts for H4.0. Since smart operations and smart working focus on different digitalization objects – the first focuses on the process and the latter on people – do organizational aspects like strategy and resources play the same role for both? And are the outcomes for both different when healthcare organizations are closer to external healthcare actors that can help with H4.0 development? The literature has not yet considered these organizational elements connected with the implementation of digital technologies for H4.0 regarding smart operations and smart working. Also, it is critical to understand the role and need of network embeddedness in H4.0, as hospitals are naturally geographically dispersed and not always close to external related actors. In such cases, the digital-resources alignment

may need a different approach to deal with. Therefore, we propose the following research questions: *What is the role of digital strategy and digital resources – as organizational strategic alignment – in the development of Healthcare 4.0 sociotechnical dimensions, namely smart healthcare operations and smart healthcare working? And what is the effect of healthcare network embeddedness in this context?*

To answer these questions, we conducted a quantitative survey involving 303 hospitals. We consider the main elements of digital strategies in organizations and the development of internal and external digital resources as antecedents of both smart operations (digital technologies for healthcare processes) and smart working (digital technologies to support workers). Furthermore, we analyze the moderating role of healthcare network embeddedness in the strategy-resources relationship. To investigate these relationships, we use hierarchical regression and bootstrapping techniques. Our results show that a stronger healthcare digital strategy positively affects the adoption of digital technologies for healthcare smart operations and healthcare smart working. On the other hand, the development of internal and external digital resources for H4.0 positively mediates only between digital strategy and smart working but there is no evidence for the same mediation in smart operations. We argue that this differentiates the focus between process automation (smart operations) and people support (smart working) in healthcare organizations. Smart working demands a stronger resource structure to embrace workers and reduce tensions related to the use of new technologies, but smart operations is more technical and less dependent on resource management, needing only a clear strategic focus for its implementation. Additionally, we found that healthcare network embeddedness positively moderates only the relationship between healthcare digital strategy and external digital resources. This indicates that a hospital's digital strategy is more effectively oriented toward leveraging external resources when it is more embedded in a network of external actors, who help clarify how such resources can be accessed and utilized. In contrast, we found no evidence that network embeddedness enhances the alignment between strategy and internal digital resources. These findings suggest that supporting smart working through external digital resources is particularly complex, as it depends not only on the availability of these resources but also on the hospital's contextual embeddedness – that is, having the necessary external connections to shape and operationalize a focused digital strategy for external resource use. Therefore, as main contribution, our study provides the theoretical basis for structuring the organizational aspects of digital transformation in healthcare by defining the main elements that consolidate a digital strategy and digital resources in H4.0. Our findings also contribute to the understanding of how to align the strategy with resources to support the implementation of digital technologies in sociotechnical healthcare systems. Healthcare managers can use these findings to understand how to create a holistic system of digital transformation aiming to build a Healthcare 4.0 organization.

2. Theoretical background

2.1. Healthcare 4.0 and the strategic alignment of sociotechnical systems

With the advent of the Fourth Industrial Revolution, also known as Industry 4.0, new information and communication technologies (ICTs) have been incorporated into organizations to digitally transform environments, enabling greater efficiency and flexibility in processes, services, and products ([Liao et al., 2017](#); [Aceto et al., 2020](#)). Due to its potential, I4.0 has expanded into other sectors, including healthcare, denoted as Healthcare 4.0 (H4.0) ([Tortorella et al., 2022a](#)). H4.0 adapts the principles and applications of Industry 4.0 to the healthcare sector, allowing real-time personalization of patient care services, professionals, and caregivers, facilitating the transition to a patient-centered environment ([Thuemmler and Bai, 2017](#)).

H4.0 has been predominantly considered from the technical point of

view with the analysis of digital technologies to increase healthcare performance (Tortorella et al., 2022a, 2022b). However, a growing emphasis has been given to the development of sociotechnical systems in healthcare organizations (Grover et al., 2018). The sociotechnical view is based on the premise that for an organization or a workplace to achieve its goals, the social and technical systems of the organization need to be interconnected and treated together (Clegg, 2000). In this sense, the literature has highlighted that digital transformation involves the integration of processes, people, and technologies in an interplay of sociotechnical elements (Roscoe et al., 2019). In such interplay, digital technologies not only modify processes but also the way human work is performed and how people interact with them to carry out their tasks (Stevens and Schaik, 2020; Zimmer et al., 2023). Therefore, when H4.0 is considered from the perspective, it can be represented from two main dimensions of technology application: smart operations and smart working.

Healthcare *smart operations* is a concept that considers the use of digital technologies to automate operational processes of the healthcare system (Tortorella et al., 2020a, 2020b). The focus of technology in this dimension is on task execution. It aims to increase the level of interoperability and efficiency, with more integrated and flexible processes to provide greater value to healthcare services and improve patient flow, reducing the possibility of errors arising from manual operations (Wang et al., 2018; Basile et al., 2024). For example, healthcare smart operations consider the use of cloud computing to store data generated by the organization, mobile applications for automatic communication, as well as the use of enterprise ICTs, such as ERP, to integrate different departments and internal and external areas (Raimo et al., 2023; Tortorella et al., 2022b). Additionally, it includes the use of telemedicine to increase the number of remote clinical consultations (Bavafa et al., 2021; Cannavacciuolo et al., 2023), the use of electronic record systems, such as electronic medical records for patient management, and digital traceability for patient and medication tracking through RFID, QR codes, or the use of process simulation to predict demands and optimize operational processes (Cerchione et al., 2023).

On the other hand, Healthcare *smart working* considers the use of digital technologies to support healthcare worker activities. In this sense, it comprises a sociotechnical evolution of digital technologies and how they are used by healthcare workers. It is based on the concept of industrial smart working proposed by Frank et al. (2019) and Cagliano et al. (2019). Healthcare smart working aims to increase ergonomics and productivity through the interplay between the worker and technology (Dornelles et al., 2022). The concept goes beyond the operational efficiency of the organization, as it also seeks to improve the well-being, health, and safety of workers, enabling the enhancement or creation of new capabilities and skills (Stevens and Schaik, 2020; Klein and Watson-Manheim, 2021; Oliveira et al., 2024). From a technology perspective, it considers the use of wireless or biomedical/digital sensors, wearable devices, big data/analytics, artificial intelligence, augmented or virtual reality, or collaborative robots to assist workers in their tasks (Wang et al., 2018; Mukherjee and Sinha, 2020; Tortorella et al., 2022b; Raimo et al., 2023). These workers include nurses, doctors, healthcare technicians, and pharmacists, who work in a diversity of areas such as emergency care, outpatient clinics, hospital wards, pharmacies, or operating rooms, dealing directly with patient care throughout the entire process of diagnosis, treatment, and follow-up (Tortorella et al., 2022b).

Strategic alignment plays an important role in the design of sociotechnical systems, as this alignment is related to the organizational structure that supports sociotechnical dimensions (Marcon et al., 2022). Such alignment has been considered a matter of establishing strategies, managing resources, and defining the form of how both are combined (Chanias et al., 2019; Verhoef et al., 2021). In the context of digital transformation, the strategy considers a combination of ICT and business model goals (Chanias et al., 2019). The healthcare digital strategy can rely on digital resources to drive the digitization of organizations

(Constantiou and Kallinikos, 2015; Vial, 2019). Digital resources include hardware, software, network, and data analytics resources (e.g., code programming or data science), which deal with the collection, storage, retrieval, sharing, and use of data, as well as financial resources for investments in basic infrastructure. Furthermore, they include related managerial domains, such as new capabilities, which typically reside in human, informational, or organizational capital, and new digital skills, both supported by a digital culture (Fernandez-Vidal et al., 2022). The organizational structure can adopt different models to manage these digital resources, implemented according to the organization's capacity and ownership (Franco et al., 2024; Correia and Frank, 2025). For example, resources can be developed *internally* or *outsourced* to external partners (Parmigiani, 2007; Chaudhuri et al., 2023; Franco et al., 2024). All these elements when combined will consolidate a H4.0 organizational system. Thus, the strategic alignment between digital strategy and resources, and the development of healthcare smart operations and smart working will define a form of how digital transformation can be faced in healthcare organizations to create a balanced system in the digital context of organizations.

2.2. Network embeddedness theory

The Network Embeddedness Theory (NET) explains that the economic and organizational activities and decisions of firms are deeply rooted in the social networks in which they are embedded (Granovetter, 1985). The term embeddedness refers to the extent to which actors' actions are embedded in networks of relationships (Granovetter, 1985). These networks—or influential intermediaries—facilitate faster and more qualified access to information, resources, and control mechanisms, which become available through the relational and structural embedding of organizations (Granovetter, 1985; Dahlke et al., 2024; Chen et al., 2025). Relational embeddedness involves direct ties between actors, considering the frequency of interaction and reciprocity, which can enhance loyalty and reduce transaction costs. Structural embeddedness, in turn, goes beyond relational ties by emphasizing that the transfer of information and/or resources in social networks does not depend solely on direct contacts, but also on the overall network structure, including indirect connections with actors close to the organization (Chen et al., 2025). Furthermore, structural embeddedness refers to an actor's position and proximity within the network, and how this influences their behavior, access to information, resources, and power (Granovetter, 1985). In addition to these elements, NET considers trust to be an essential operational component. Since institutional arrangements alone are not sufficient to fully prevent opportunistic behavior or fraud, trust among network actors often allows formal contracts to be replaced, thereby reducing the risks associated with partnerships (Granovetter, 1985). Therefore, network embeddedness can be a mechanism through which organizations seek to adapt to or align with the external environment and acquire essential resources for specific purposes (Gnyawali and Madhavan, 2001).

Recent studies in the field of digital transformation, such as Chen et al. (2024), highlight the role of network embeddedness in enabling digital capabilities through technologies aimed at production customization. Similarly, Chen et al. (2025) demonstrated the contribution of network embeddedness in supply chains to strengthening operational resilience through digitalization. In addition, other studies, such as Lyu et al. (2020), identified the positive effects of network embeddedness on the relationship between open innovation and innovation radicalness, while Xing et al. (2023) proposed an approach using the TOE framework, positioning network embeddedness as an environmental-level factor that supports digital transformation and enhances innovation performance in manufacturing. Dahlke et al. (2024) also found evidence that the adoption of artificial intelligence is related to three epidemic effect mechanisms of network embeddedness. Collectively, these studies demonstrate how network embeddedness can support both digital transformation and organizational performance and innovation.

Therefore, network embeddedness can be seen as an additional external dimension of sociotechnical systems applied to DT (Xing et al., 2023). This is because, beyond emphasizing the importance of organizations being embedded in social networks, this perspective expands the understanding of DT by incorporating the external environment as a sociotechnical dimension (Marcon et al., 2022; Xing et al., 2023; Marcon et al., 2025). However, the literature has not yet explored how the degree of network embeddedness can support digital transformation in the healthcare context, especially when viewed as a socio-technical system.

In this context, the structuring of H4.0 in health systems requires an appropriate environment to support its implementation, as highlighted by the sociotechnical literature (Marcon et al., 2022; Xing et al., 2023; Marcon et al., 2024), which also emphasizes the importance of external context and participation in the digitalization process. As noted in the previous section, from a technical standpoint, this includes the dimension of healthcare smart operations; from the social perspective, healthcare smart working; and from the organizational side, the alignment between digital strategy and digital resources. However, it is evident that the external environment of healthcare systems still needs to be incorporated to fully realize the concept of a sociotechnical system—integrating social, technical, organizational, and environmental aspects (Marcon et al., 2025, 2024). In this regard, network embeddedness can serve as a reliable approach to acting as the environmental dimension of H4.0, as it can facilitate the alignment between strategy and digital resources by enabling healthcare systems to access essential digital information and resources—through direct or indirect connections with various network actors—to support the implementation of the smart dimensions of H4.0.

3. Hypotheses development

3.1. Healthcare digital strategies and the development of Healthcare 4.0 dimensions

The implementation of digital technologies demands a strategic definition to set goals and priorities of which technologies should be applied and how they must support processes and people (Enrique et al., 2023). From an organizational perspective, a digital strategy needs to establish digital goals for the organizational processes that will be changed with the support of digital technologies (Vial, 2019). When a sociotechnical system is aimed to be supported by digital technologies, the consolidation of this digital environment with clear rules will become the organizational driver of the digitalization process (Marcon et al., 2022; Marcon et al., 2025). Without a clear digitalization strategy, technologies tend to be implemented to solve local problems, but at the risk of missing the big organizational picture that addresses the systemic problems through digital solutions (Baiyere et al., 2020).

When it comes to H4.0, the development of a digital strategy should be a priority for the implementation of both healthcare smart operations and smart working. First, smart operations consider digital technologies applied to healthcare processes, but which processes are there? Of course, we can refer to the whole healthcare system and its processes, but these are too many and too complex to consider ideally a complete automated system. Healthcare is a sociotechnical organization especially because it is embedded in human-related activities that are performed based on intensive use of high-skilled knowledge (Beane, 2019; Stevens and Schaik, 2020). Therefore, it cannot be simply automated—the digital strategy would be the organizational roadmap to define priorities and needs regarding the technologies that should be implemented to address processes that are the aim of digitalization (Baptista et al., 2020; Fischer et al., 2020). Another aspect is the competition for resources: a digital strategy creates financial priorities to invest in digital technologies of the healthcare organization; otherwise, a corporate strategy without digitalization as a priority will dilute financial resources in many other demands (Piccoli et al., 2024). Furthermore, it is important to consider that the digital strategy can also have as an aim

the benefit of healthcare workers, especially in healthcare systems where workers' resilience is necessary and technologies can play an important role in it (Tortorella et al., 2022c). A clear digital strategy can set workers as a keystone of the sociotechnical healthcare system and define the development of smart working as a priority, as it also happens in industrial contexts (Marcon et al., 2022). Therefore, when the healthcare digital strategy is positively associated with both smart operations and smart working, it can be considered a strategic system of the organization that has a clear vision of creating a sociotechnical balance. We hypothesize this as follows:

H1. The implementation of an organizational digital strategy is positively associated with higher levels of adoption of healthcare 4.0 technologies.

H1a. The implementation of a digital strategy is positively associated with higher levels of healthcare smart operations (use of technologies to automate healthcare processes).

H1b. The implementation of a digital strategy is positively associated with higher levels of healthcare smart working (use of technologies to support healthcare workers).

3.2. The alignment of healthcare digital strategy and resources and the role of network embeddedness

Verhoef et al. (2021) describe strategic imperatives of digital transformation, considering digital resources as an element to be combined with the strategy and the goal of the digitalization process. This line of research establishes that for a digital transformation strategy to be developed, it needs to consider digital resources as a central element of its digital technology architecture (Piccoli et al., 2024). It means that digital transformation is not just about defining what to digitalize (goals) and adopting the technologies (as considered in H1). Instead, a set of organizational digital resources needs to be established to provide support to the whole sociotechnical system. These digital resources are necessary to act as support for the digitalization process which tends to be challenging due to several organizational barriers that need to be overcome during the digitalization (Verhoef et al., 2021). In this sense, a strategic alignment is necessary between the digital strategy and the resources to manage the digitalization processes (Fischer et al., 2020). Some authors have considered the digital strategy as a bundle of goals, organizational structure, and resources (Verhoef et al., 2021; Fischer et al., 2020), but this may be a consequence of a qualitative analysis of how companies establish the strategy and organizational requirements. We follow a classic stream of information systems that differentiates between strategy and resources and consider strategy as the antecedent for the development of the resources (Kearns and Lederer, 2000; Yeow et al., 2018). According to this view, both are independent but interrelated elements of organizational development (Yeow et al., 2018). Furthermore, these resources, which can comprise a vast type of elements including financial, human, and infrastructure factors (Verhoef et al., 2021; Franco et al., 2024), can be part of the established internal digital transformation or can be acquired from external partners, as a make-or-buy condition (Parmigiani, 2007; Franco et al., 2024). This is more evident in the digital era, as organizations do not master the whole set of digital resources necessary to develop the digitalization process (Dalenogare et al., 2023). This is the case of healthcare systems, which are not essentially born-digital. Instead, healthcare organizations need to set new strategies for the digital transformation process (Denicolai and Previtali, 2023). Consequently, healthcare organizations will also need to build and internalize digital resources that are supported by the strategy and that can help to implement the digitalization goals (Denicolai and Previtali, 2023). Therefore, we establish the following hypothesis:

H2. The implementation of an organizational digital strategy is positively associated with higher levels of development of digital resources.

H2a. The implementation of a digital strategy is positively associated with higher levels of internal digital resources.

H2b. The implementation of a digital strategy is positively associated with higher levels of external digital resources.

Network embeddedness suggests that the position an organization occupies within a network influences its access to information and resources (Lin, 2001). This occurs because such information and resources are transmitted or reinforced through a chain of interconnected nodes—where each node represents an external actor—and the proximity and degree of connection among these actors facilitate the receiving organization's ability to understand its strategy and focus on resource development (Granovetter, 1985; Mazzola et al., 2015). Network embeddedness requires organizations to actively locate, process, and interact with different actors in the network that are geographically close and can provide the required information and resources (Dahlke et al., 2024). Geographic proximity refers to the physical distance between organizations, which can encourage more frequent interactions and, consequently, foster the formation of stronger healthcare environments. Being physically and cognitively close to these sources can facilitate the understanding of the best pathways for digital transformation, helping to the formulation of more effective strategies aligned with the necessary resources for the digitalization of healthcare (Chen et al., 2025).

On the one hand, geographic proximity to external network actors can help an organization develop an outward-looking perspective, fostering learning from other actors and based on that, formulating a digital strategy aimed at strengthening its internal digital resources. This occurs because, by being embedded in a network, the organization has the opportunity to observe which digital resources are being used by other actors—or which are necessary to support digitalization—and based on this learning, develop and manage such resources internally (Miroshnichenko et al., 2021; Xing et al., 2023; Chen et al., 2025). For example, a healthcare organization may capture knowledge about the development of technological solutions from another organization, whether public or private, within the same network and apply that knowledge internally by training its professionals to create an innovative and customized solution (Miroshnichenko et al., 2021). Alternatively, it may establish partnerships with universities and research institutions to access information and experiences related to digital transformation.

On the other hand, proximity to external network actors can assist an organization (e.g., a hospital) in identifying the most suitable partners for full access to external resources, enabling effective strategies to obtain those resources (Granovetter, 1985). This is because network embeddedness demonstrates that an organization can observe the trust levels and potential shortcomings among its potential suppliers—allowing for better comparisons—evaluating competitive advantages and exercising greater flexibility in accessing information and resources (Granovetter, 1985). For example, by occupying a central position in a network, an organization may gain access to suppliers offering more innovative solutions within its geographic context (Chen et al., 2025); it may also engage in collaborative environments that enable mechanisms for knowledge transfer and digital technologies (Lyu et al., 2020; Chen et al., 2024); or even establish contact with healthcare operators (such as insurers, health plans, and provider networks), whose regional commercial demands can drive the development of new digital healthcare solutions.

In this context, it is observed that the success of implementing H4.0 in hospitals may depend precisely on the integration between strategy, digital resources, and digital network embeddedness. By integrating these dimensions, health systems can define organizational structures for healthcare digital transformation that go beyond the hospital's internal walls, also considering how external factors influence access to resources and, at the same time, how they can be influenced by it (Correia and Frank, 2025). The key point is that, by integrating these

three dimensions, hospitals can not only achieve internal digitalization but also develop the capability to orchestrate digital transformation within health innovation ecosystems, bringing together resource providers to align digital solutions with the hospital's needs and, simultaneously, supporting network partners in their digitalization (Correia and Frank, 2025), thus fostering multiple value gains for all stakeholders involved.

Based on these arguments, network embeddedness plays a crucial role in the digital strategy-resources alignment, as it provides a source of information and resources which is important to develop the digital transformation of healthcare activities better. Thus, we propose the following hypothesis:

H3. The level of network embeddedness hospital can have in a healthcare ecosystem will positively moderate and strength the alignment between digital strategy and resources.

H3a. The level of network embeddedness positively moderates the relationship between digital strategy and internal digital resources.

H3b. The level of network embeddedness positively moderates the relationship between digital strategy and external digital resources.

3.3. Contribution of digital resources for the development of Healthcare 4.0 dimensions

A good strategy-resource alignment should help to develop the organizational information system for digital transformation (Yeow et al., 2018). Prior research on digital transformation has defended the need for structuring also digital resources to enhance technological implementation in the digital transformation process (Liu et al., 2011; Yeow et al., 2018). In the H4.0 perspective, this means that digital resources, when aligned to the digital strategy, should support the development of the sociotechnical H4.0 system. It is like what was previously argued by Marcon et al. (2022) and Marcon et al. (2024) in industrial systems, where the combination of organizational strategies and resources (sociotechnical environment) provided the basis for Industry 4.0 technologies implementation. In this same vein, digital resources should provide the basis for the development of smart operations and working technologies. For instance, digital resources include the organizational culture and informational structure necessary to implement digital technologies in processes and change the form such processes are executed (Fischer et al., 2020; Marcon et al., 2022; Gillani et al., 2024; Marcon et al., 2025), as proposed for healthcare smart operations. Also, digital resources include skills and capabilities necessary to support people, when they need to learn how to use new digital technologies for their task execution, as considered in healthcare smart working (Verhoef et al., 2021; Fernandez-Vidal et al., 2022; Marcon et al., 2025, 2024). Additionally, digital resources can be managed internally and externally, as these are totally different sources and can require different strategic management approaches to deal with (Dalenogare et al., 2023). Consequently, they can have different contributions to smart operations and smart working. External digital resources can consider the orchestration of knowledge and new skills through collaboration that can change the way business processes are executed, affecting smart operations. It also can create new mindsets in workers helping to shape the way technologies are used in healthcare activities (Fischer et al., 2020). On the other hand, internal digital resources can help consolidate systems legacy and information systems teams and knowledge that can provide a basis for automating processes through digital technologies (Liu et al., 2011) and for expanding workers skills to the use of new digital technologies (Verhoef et al., 2021). Consequently, these different arguments point to a mediating role of healthcare digital internal and external resources in the strategic alignment for the development of an H4.0 sociotechnical system. We summarize this in the following hypotheses H3 (for internal resources) and H4 (for external resources):

H4. The development of internal healthcare digital resources is positively associated with higher levels of adoption of healthcare 4.0 technologies.

H4a. The development of internal healthcare digital resources is positively associated with higher levels of healthcare smart operations (use of technologies to automate healthcare processes).

H4b. The development of internal healthcare digital resources is positively associated with higher levels of healthcare smart working (use of technologies to support healthcare workers).

H5. : The development of external healthcare digital resources is positively associated with higher levels of adoption of healthcare 4.0 technologies.

H5a. The development of external healthcare digital resources is positively associated with higher levels of healthcare smart operations (use of technologies to automate healthcare processes).

H5b. The development of external healthcare digital resources is positively associated with higher levels of healthcare smart working (use of technologies to support healthcare workers).

All the hypotheses presented in this section are summarized in the conceptual model of Fig. 1. This figure illustrates, in its first part, the strategic alignment between healthcare digital strategy and the management of digital healthcare resources (both internal and external) and the moderating role of healthcare network embeddedness. Then, in its second part, Fig. 1 describes the mediating role of healthcare digital resources between healthcare digital strategy and the dimensions of Healthcare 4.0, namely smart operations and smart working.

4. Research method

4.1. Sampling

We focused our research unit on hospital management. Therefore, our target respondents are middle and senior-level healthcare leaders (directors, managers, or supervisors) who deal with healthcare technologies for hospital management. Also, we focused on internal elements of the hospitals, like the level of development of digital strategies and resources and the level of adoption of digital technologies. Therefore, our data source was the single respondent, as secondary data is not available for this research aim and multiple respondents would not be able to respond to issues related to digital transformation, which are usually under the coordination of technology/operations managers

(Frank et al., 2019). This should be less biased than selecting experts from specific departments to analyze the system (Ketokivi, 2019). We defined the population for our sample based on the Annual Report on the Hospital Scenario in Brazil, developed by the Brazilian Hospital Federation (FBH). To mitigate the socioeconomic effect, we restricted the sample of our study to Brazilian hospitals. Although this criterion may limit the generalization of our findings, it strengthens the internal validity of our results. The population consists of 7242 hospitals registered in the database of the National Registry of Health Establishments (CNES), DataSUS, and the Ministry of Health (MS). Considering a 5 % error and a 90 % confidence interval, it was determined that the minimum sample size needed to research hospitals and achieve national representativeness is 263. From this population, the initial sample available with the target contacts consisted of 3823 hospitals from all regions of Brazil.

Data collection was carried out through three channels. Firstly, the survey was emailed to as many hospitals as possible identified in the sample. Secondly, a search for the sampled hospitals was conducted on the LinkedIn platform. Thirdly, the survey was manually delivered in printed forms. Three waves of the questionnaire were sent over three months. As a result, 316 initial responses were obtained; after excluding incomplete and duplicate questionnaires, we obtained 303 considered valid (response rate of 8 %) for this research. The final sample size can be considered representative of Brazilian hospitals, as it exceeds the minimum requirement of 263 (90 % confidence and 5 % error), representing the proportions of the population. Table 1 presents the study sample profile.

Table 1
Sample composition ($n = 303$ hospitals).

Number of beds	(%)	Respondent role	(%)
≤ 150	33 %	Middle level (e.g., supervisors)	82.18 %
> 150	67 %	Senior level (e.g., manager or director)	17.82 %
Number of employees		Hospital age	
1–100	12 %	< 20	4.0 %
101–200	14 %	20–50	49.50 %
201–500	110 %	> 50	46.50 %
501–1000	115 %	Hospital's profile	
1001–3000	36 %	Teaching	10.89 %
> 3000	16 %	Public	36.30 %
		Private	52.81 %

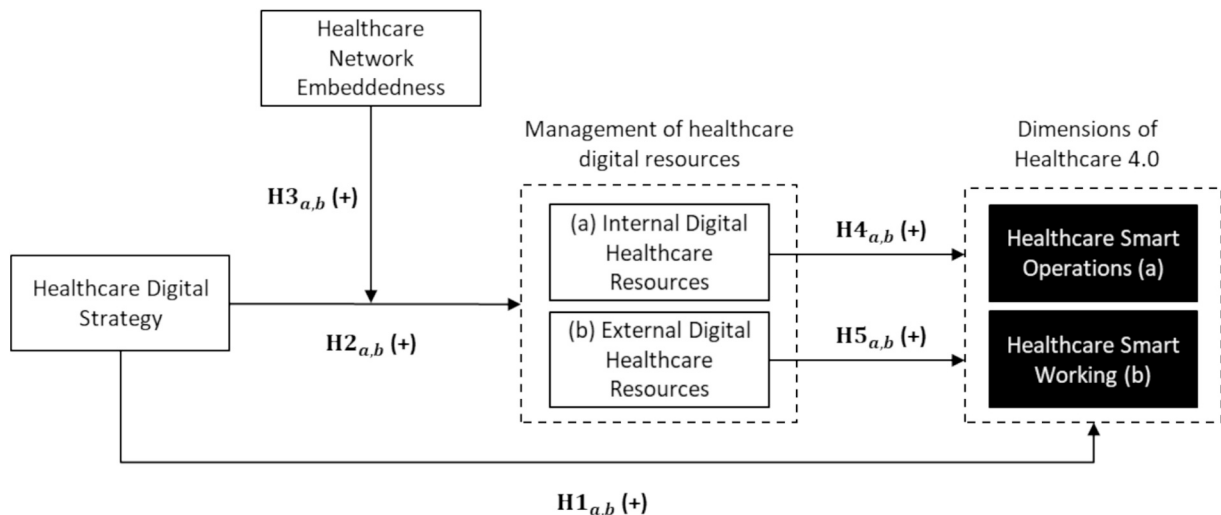


Fig. 1. Conceptual research model.

4.2. Survey instrumentation

Our research instrument was developed based on constructs from the literature and adapted to the specific objectives of our study. The proposed instrument consists of six blocks of questions (see Appendix A): (i) the level of implementation of healthcare digital strategies (STRATEGY), adapted from [Nasiri et al. \(2020\)](#) and [Verhoef et al. \(2021\)](#); (ii) the level of ownership of Internal Digital Healthcare Resources (INTERNAL), adapted from [Barney \(1986, 1991\)](#) and [Franco et al. \(2024\)](#); (iii) the level of outsourcing of External Digital Healthcare Resources (EXTERNAL), adapted from [Dyer and Singh \(1998\)](#); [Dyer et al. \(2018\)](#), [Franco et al. \(2024\)](#); (iv) the level of healthcare network embeddedness (EMBEDDEDNESS), adapted from [Xing et al. \(2023\)](#), [Chen et al. \(2024\)](#), [Chen et al. \(2025\)](#) (v) the level of implementation of H4.0 digital technologies in healthcare management activities, proposed by [Frank et al. \(2019\)](#), [Dornelles et al. \(2022\)](#), [Tortorella et al. \(2022a, 2022b\)](#) and [Deloitte \(2020\)](#); (vi) information about the interviewees and their hospitals was collected to determine the demographic profile of the sample. Respondent information (e.g., role) was used to verify the legitimacy of their opinions and ensure a broader understanding of the investigated phenomenon. Meanwhile, hospital information allowed for controlling some contextual characteristics (e.g., number of beds, number of employees, hospital age in operation years, and ownership) that may influence the healthcare digital transformation process ([Raimo et al., 2023](#)).

To ensure the feasibility of survey administration, the instrument items were assessed and pre-tested by fifteen randomly selected experts (three academics and twelve healthcare practitioners) to ensure the face and content validity of the scale items ([Taherdoost, 2016](#)). The experts recommended minor terminological changes to enhance the clarity and comprehensibility of the survey items. The questions considered in this study were measured on a five-point Likert scale, ranging from 1 (strongly disagree) to 5 (strongly agree) for evaluating blocks (i), (ii), (iii), and (iv) and from 1 (never) to 5 (very frequently) to assess the usage of digital technologies in sub-block (v).

4.3. Constructs definition and validation

H4.0 digital technologies are typically organized into bundles of applications representing formative constructs (e.g., [Tortorella et al., 2022a](#)). We collected the information regarding the technologies usage through the questionnaire without using predefined bundles, as they were determined after through the use of Exploratory Factor Analysis (EFA). We conducted the EFA using orthogonal Varimax rotation to minimize potential ambiguities in the distribution of variables ([Hair et al., 2018](#)). We selected those factors with eigenvalues >1 ([Hair et al., 2018](#)), representing 67.953 % of the total variance in responses, as shown in [Table 2](#). We also assessed parameters validating the use of EFA for our selected dataset. These parameters include the Kaiser-Meyer-Olkin (KMO) test (which should be >0.500), Bartlett's sphericity test (which should be <0.05), the anti-image matrix for Measures of Sampling Adequacy (MSA) (which should be >0.500), and the percentage of explained variance (which should be >60 %). All our obtained parameters were in line with the suggested thresholds in the literature (KMO = 0.930; Bartlett <0.000 ; MSA ≥ 0.879 ; % of variance = 67.953 %) ([Hair et al., 2018](#)).

Analyzing the results of the factor loadings in each component, we labeled the two digital application bundles based on the composition of technologies and their predominant focus, following a procedure similar to previous literature ([Tortorella et al., 2022b](#)). As a result: Healthcare Smart Working (WORK) and Healthcare Smart Operations (OPERATIONS). [Table 2](#) also displays Cronbach's alpha coefficients for these constructs obtained from the EFA, confirming reliability with values exceeding 0.9. Subsequently, we used the factorial scores of the two main factors obtained through the Thurstone method. More specifically, standardized scores were used in the subsequent linear regression

Table 2

Exploratory Factor Analysis for digital technologies.

Variables	WORK	OPERATIONS	Comm.
Big Data/analytics	<u>0.742</u>	0.448	0.752
Artificial intelligence/machine learning	<u>0.763</u>	0.420	0.759
Blockchain	<u>0.828</u>	0.169	0.714
Augmented and virtual reality	<u>0.717</u>	0.206	0.557
Collaborative robotics (e.g., surgical robots, cobots, outpatient care, AGVs, AMRs, etc.)	<u>0.649</u>	0.448	0.622
Biomedical/digital wireless sensors	<u>0.746</u>	0.259	0.623
Digital patient relationship applications	<u>0.630</u>	0.433	0.584
Cloud Computing	0.539	<u>0.621</u>	0.677
Mobile apps	0.456	<u>0.619</u>	0.590
Corporate management systems (e.g., ERP - Enterprise Resource Planning)	0.250	<u>0.844</u>	0.774
Electronic health records (e.g., electronic prescription, electronic medical record)	0.146	<u>0.871</u>	0.780
Digital traceability (e.g., RFID, QR Code, etc.)	0.371	<u>0.790</u>	0.761
Simulation to support activities	0.502	<u>0.646</u>	0.669
Telemedicine	0.540	<u>0.599</u>	0.651
Extraction sums of squared loadings	8.355	1.158	
% of variance explained	59.682	8.271	
% of variance explained (cumulative)	59.682	67.953	
Rotation sums of squared loadings	4.971	4.543	
% of variance explained	35.504	32.449	
% of variance explained (cumulative)	35.504	67.953	
Cronbach's Alpha	0.906	0.918	

Note: Factor loadings greater than 0.50 are underlined and bolded.

analysis.

For the evaluation of healthcare digital strategy (STRATEGY), healthcare digital resources (INTERNAL and EXTERNAL) and healthcare network embeddedness (EMBEDDEDNESS), we conducted a Confirmatory Factor Analysis (CFA) on constructs adapted from the literature. As indicated by the Comparative Fit Index (CFI), Tucker-Lewis Index (TLI), Root Mean Square Error of Approximation (RMSEA), Average Variance Extracted (AVE), Composite Reliability (CR), and Cronbach's Alpha, the analyzed constructs exhibited a good fit. All these values were considered acceptable and most exceeded predefined thresholds ([Hair et al., 2018](#)), as presented in [Table 3](#). Additionally, the items demonstrated high factor loadings for each construct (Appendix A). We also ensured discriminant validity by applying the criteria of [Fornell and Larcker \(1981\)](#), which involve assessing correlation values with the square root of the AVE for each construct. Through this evaluation, we confirmed that the constructs built through the CFA procedure met the criteria proposed by [Fornell and Larcker \(1981\)](#). Additionally, we analyzed the means, standard deviations, and correlations between our constructs and the control variables incorporated into the model. We considered four control variables based on previous studies ([Tortorella et al., 2022b](#); [Raimo et al., 2023](#)), namely: number of beds, number of employees, and two dummy variables, the first indicating whether the hospital is teaching, and the second indicating whether the hospital is public. Bed and employee data were logarithmically transformed for inclusion in our regression model. [Table 4](#) presents the descriptive statistics, Fornell-Larcker criteria, and bivariate correlations.

4.4. Response bias and common method variance

To mitigate common method variance (CMV), we conducted a pre-test of the instrument's questions with a randomized subsample drawn from our target population. For this purpose, we pretested the questionnaire to ensure the clarity of our research instrument (as explained in [Section 3.2](#)). Additionally, we randomized the order of variables and questionnaire blocks to enhance their reliability and prevent order bias and primacy bias ([Podsakoff et al., 2003](#)). As pointed out by [Ketokivi \(2019\)](#), in surveys involving questionnaires, it is impossible to eliminate bias in the research. However, we implemented reasonable countermeasures to address this concern in our study.

Table 3

Results of confirmatory factor analysis.

Construct	CFI	TLI	RMSEA	AVE	CR	Cronbach's α
[STRATEGY]	0,990	0,983	0,070	0,685	0,928	0,928
[INTERNAL]	0,995	0,989	0,043	0,506	0,835	0,831
[EXTERNAL]	0,968	0,947	0,080	0,444	0,826	0,823
[EMBEDDEDNESS]	0,987	0,979	0,072	0,637	0,913	0,913

Table 4

Bivariate correlations.

Variables	1	2	3	4	5	6	7	8	9	10
1 Ln_beds	–									
2 Ln_employees	0.849**	–								
3 Dummy_teaching	0.183**	0.262**	–							
4 Dummy_public	0.189**	0.156**	–0.082	–						
5 STRATEGY	0.007	0.077	0.193**	–0.579**	0.830					
6 INTERNAL	0.013	0.024	0.189**	–0.461**	0.676**	0.711				
7 EXTERNAL	0.054	0.074	0.100	–0.174**	0.358**	0.529**	0.667			
8 EMBEDDEDNESS	0.147*	0.185**	0.140*	–0.260**	0.556**	0.508**	0.386**	0.800		
9 WORK	–0.087	–0.138*	0.188**	–0.232**	0.295**	0.430**	0.360**	0.378**	–	
10 OPERATIONS	0.148*	0.261**	0.218**	–0.515**	0.628**	0.491**	0.269**	0.526**	0.000	–
Mean	5.347	6.423	0.116	0.396	3.448	2.807	2.937	3.049	0	0
S.D.	0.701	0.881	0.320	0.490	1.025	0.895	0.745	1.008	1	1

OBS1: ** $p < 0.01$; * $p < 0.05$. OBS2: Bold numbers in the diagonal report Fornell-Lacker criteria for discriminant validity.

4.5. Data analysis

To test our hypotheses, we conducted a series of hierarchical ordinary least squares (OLS) regressions. Before the regression analysis, we standardized the variables using a mean-centering Z-score. We employed a bootstrap approach based on the 'Process' macro proposed by Hayes (2017) to check for indirect mediation effects. To test the moderation effects (H3a and H3b), we standardized both the independent and moderating variables using mean-centering Z-scores and multiplied the moderator by the independent variable. This allowed us to create a multiplicative score to assess the interaction effect.

Before focusing on the interpretation of the results, we conducted a series of post hoc analyses to verify if all the conditions for OLS regression were met. Firstly, we assessed the normality condition required for OLS regression, employing three strategies based on the analysis of residual normality: we examined the density plot (histogram), the standardized normal probability (P–P) plot, and the quantile plot of the variable against the quantiles of a normal distribution. All these plots indicated a normal distribution of residuals for our OLS regression analysis. No model rejected the hypothesis of adherence to the normal distribution of residuals (Kolmogorov-Smirnov test p -value < 0.01). Secondly, we checked linearity through partial regressions for independent variables. Thirdly, we also assessed the homoscedasticity requirements for OLS using visual analysis with standardized residuals plotted against predicted values. Lastly, we examined the potential issue of multicollinearity in the OLS model through the Variance Inflation Factor (VIF). The results showed low multicollinearity, indicating a low VIF (< 3.847) for all independent variables, well below the threshold of 10 (Hair et al., 2018). Additionally, we examined the potential correlation of standardized residuals with independent variables and found no pattern suggesting multicollinearity.

5. Results

The results of the hierarchical regression are presented in Table 5. In the first main stage, hierarchical regression models were executed in four steps. In the first step, we included only the control variables. In the second step, we included STRATEGY in both sources of digital healthcare resources (INTERNAL and EXTERNAL). In the third step, we add EMBEDDEDNESS, and in the fourth step, we test the moderation effect of EMBEDDEDNESS $\times$ STRATEGY on both sources of digital healthcare

resources (INTERNAL and EXTERNAL). In the second main stage, we tested the direct effect of STRATEGY on WORK and OPERATIONS (initially including only the control variables and then including the independent variable STRATEGY). Finally, in the third main stage, hierarchical regression models were executed, including the independent variable STRATEGY and the mediator's variables INTERNAL and EXTERNAL.

As shown in Table 5, the final model (3rd main stage) was statistically significant in both WORK ($F = 15.263$; $p < 0.01$) and OPERATIONS ($F = 43.986$; $p < 0.01$), explaining 24.8 % and 49.9 % of the variance (Adjusted R^2), respectively. For the first main stage, regarding STRATEGY, our results showed a statistical association with INTERNAL ($B = 0.611$; $p < 0.000$) (H2a is supported) and EXTERNAL ($B = 0.375$; $p < 0.000$) (H2b is supported). The moderation model for EMBEDDEDNESS between STRATEGY and INTERNAL ($F = 42.537$; $p < 0.01$) explains 49.1 % of the variance (adjusted $R^2 = 0.491$), but we found no statistical evidence supporting this moderation (H3a is not supported). On the other hand, the moderation model for EMBEDDEDNESS between STRATEGY and EXTERNAL ($F = 12.468$; $p < 0.01$) explains 21.0 % of the variance (adjusted $R^2 = 0.210$). We observed a positive moderation effect between STRATEGY and EMBEDDEDNESS ($B = 0.231$, $p < 0.000$) (H3b is supported). To better understand the effect of this moderation, we plotted the slope in Fig. 2, which shows the positive effect of STRATEGY on EXTERNAL when the level of EMBEDDEDNESS is high in hospitals. In other words, when a hospital has a high level of network embeddedness within a healthcare ecosystem, it strengthens the alignment between digital strategy and digital resources, and this effect is particularly positive in the relationship between digital strategy and external digital healthcare resources.

For the second main stage, the direct effect of STRATEGY was statistically significant for WORK ($B = 0.258$; $p < 0.01$) (H1a is supported) and OPERATIONS ($B = 0.416$; $p < 0.000$) (H1b is supported). Finally, for the third main stage, the results indicate a complete mediation effect only in the relationship between STRATEGY and WORK ($F = 15.263$; $p < 0.01$). Only the INTERNAL and EXTERNAL digital healthcare resources become the only statistically significant variables in the model.

However, to assess mediation, indirect effects must be calculated (Preacher and Hayes, 2008). Therefore, we employed a procedure proposed by Hayes (2017), called PROCESS, to evaluate the mediation effects of our model. PROCESS involves a bootstrapping procedure to examine conditional indirect effects, making it more robust when

Table 5
Regression results for OLS direct effects^a.

Variables	1st main stage				2nd main stage				3rd main stage			
	INTERNAL		EXTERNAL		WORK		OPERATIONS		WORK		OPERATIONS	
	Model 1	Model 2	Model 3	Model 4	Model 1	Model 2	Model 3	Model 4	Model 1	Model 2	Model 1	Model 2
Dummy_hospital_research	0.434**	0.233	0.233	0.234	0.196	0.073	0.073	0.041	0.693**	0.608**	0.256	0.119
Dummy_hospital_public	-0.948**	-0.214*	-0.235*	-0.235*	-0.373**	0.076	0.049	0.033	-0.404*	-0.094	-0.052	-0.606**
Ln_beds	0.089	0.137*	0.121	0.121	0.012	0.042	0.021	0.070	0.180	0.200*	0.154	-0.123
Ln_employees	-0.016	-0.143	-0.157*	-0.157*	0.076	-0.002	-0.020	-0.072	-0.318*	-0.371**	-0.332**	0.373**
STRATEGY		0.611**	0.493**	0.493**	0.375**	0.219**	0.219**	0.326**		0.258**	0.014	0.352**
INTERNAL											0.273**	0.094
EXTERNAL											0.206**	0.018
EMBEDDEDNESS			0.205**	0.205**			0.268**	0.258**				
EMBEDDEDNESS x STRATEGY			-0.001	-0.001				0.231**				
F-test	23.685**	53.606**	49.795**	42.537**	3.462**	9.026**	10.818**	12.468**	9.757**	11.092**	15.263**	43.986**
R ²	0.241	0.474	0.502	0.502	0.044	0.132	0.180	0.228	0.116	0.157	0.266	0.511
Adjusted R ²	0.231	0.466	0.492	0.491	0.032	0.117	0.163	0.210	0.104	0.143	0.248	0.499
Change in R ²	0.241**	0.233**	0.028**	0.000	0.044**	0.088**	0.048**	0.048**	0.116**	0.042**	0.266**	0.511**

n = 303.

** p < 0.01.

* p < 0.05.

^a Unstandardized beta coefficients are reported, since the main variables were standardized before regression.

compared to the Sobel z-test for testing mediation effects (Zhao et al., 2010). To operationalize PROCESS, as suggested by Preacher and Hayes (2008), we configured 5000 bootstrap samples. Table 6 presents the direct effects, indirect effects (including standard deviation, significance level, and corresponding lower (LLCI) and upper (ULCI) confidence intervals), the total effect (which is given by the sum of direct and indirect effects), and the conclusion.

For OPERATIONS, the direct effect of STRATEGY was statistically significant (effect = 0.5446; $p < 0.001$). However, regarding the indirect effects, the bootstrap results indicated that for both INTERNAL and EXTERNAL, the values fall outside the 95 % confidence interval. In this case, the hypothesis of mediation for each digital resource source is rejected, as the value “0” is within the interval (LLCI and ULCI), indicating that the indirect effect may not exist. However, this situation requires further investigation to obtain more conclusive results. For WORK, the direct effect of STRATEGY was not statistically significant (effect = 0.0084; $p = 0.9041$). Regarding the indirect effects, the bootstrap results indicated significance for both INTERNAL and EXTERNAL, as both are within the 95 % confidence interval, suggesting complete mediation for WORK.

5.1. Power analysis

Based on the regression results reported in Table 5, we conducted a power analysis of the regression models using Cohen's f^2 estimation (Cohen et al., 2003). We also calculated the population effect size of R^2 using Cohen's f^2 estimation (Cohen et al., 2003) from the two final models in the last stage of our OLS hierarchical procedure. For the WORK, we obtained $f^2 = 0.362$, representing a statistical power of ≈ 1.00 at $\alpha = 0.01$. For the OPERATIONS, we obtained $f^2 = 1.045$, indicating a statistical power of ≈ 1.00 at $\alpha = 0.01$. Therefore, the result of the effect size for each model is considered an average effect for the explanatory variables (Cohen et al., 2003).

5.2. Robustness checks

We assessed the robustness of our model by applying various strategies to ensure the stability and consistency of the results. Following the approach of previous studies (e.g., Lerman et al. (2022) and Franco et al., 2023), we established an interactive protocol. First, we removed control variables and checked whether the results remained stable. The removal of control variables demonstrated that predictor variables were not artifacts of control variables. Second, we conducted an individual predictor analysis. Lastly, in the third step, we tested a competitive model, considering the healthcare digital strategy as a moderator between healthcare digital resources and the implementation of smart healthcare dimensions. However, we did not find statistical support for this competitive model. Overall, we found stable results, as there were no significant changes in the coefficients of all models altered in our hierarchical regression. All these strategies indicated a stable behavior of our proposed model.

6. Discussions and conclusions

Our results demonstrate that H4.0 encompasses a sociotechnical view through two smart dimensions, which we refer to as healthcare smart operations and healthcare smart working. The former focuses on introducing technologies for task execution, aiming to make them more automated, especially those that are inherently mechanical. The second dimension focuses on workers, who are the actors responsible for task execution but are supported by the introduction of digital technologies, enabling an upskilling effect, which means enhancing worker competence (Dornelles et al., 2023). Both dimensions can be applied across hospital units. However, despite the digital technologies addressed in this study focusing on the first or second dimension, some of them can simultaneously encompass both. A potential explanation for this is the

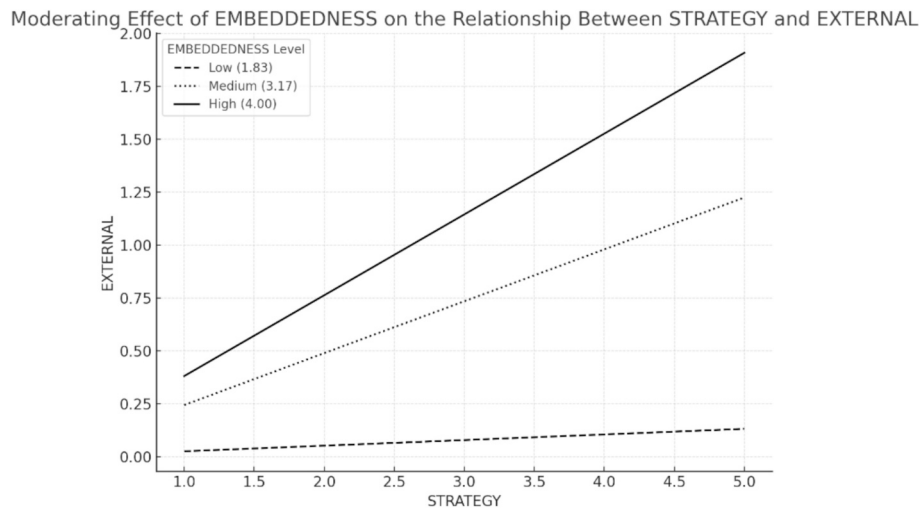


Fig. 2. Slope for the moderating role of healthcare network embeddedness (EMBEDDEDNESS) on STRATEGY vs EXTERNAL.

Table 6

Bootstrapping results of the indirect effects.

Mediation path	Direct effect		Indirect effect				Total Effect		Conclusion
	Effect	Sig.	Effect	SD	LLCI (95 %)	ULCI (95 %)	Effect	Sig.	
STRATEGY- > INTERNAL- > WORK	0.0084	0.9041	0.2205	0.0582	0.1113	0.34621	0.2951	0.0000	Complete
STRATEGY- > EXTERNAL- > WORK			0.0662	0.0240	0.0155	0.1108			Complete
STRATEGY- > INTERNAL- > OPERATIONS	0.5446	0.0000	0.0784	0.0464	-0.0107	0.1709	0.6278	0.0000	No mediation
STRATEGY- > EXTERNAL- > OPERATIONS			0.0048	0.0186	-0.0311	0.0422			No mediation

flexibility of application and the impact that technologies can have on different areas, such as telemedicine. Although our research suggests that telemedicine falls within the healthcare smart operations dimension, in practice, it also plays a significant role in the social context, as previously shown by Cannavacciuolo et al. (2023). Telemedicine is a technology that supports healthcare workers in providing real-time and virtual care, primarily in emergency situations, allowing for the acceleration of clinical procedures. Furthermore, it benefits patients by reducing waiting times for care or the need to travel to a closer clinic. Therefore, our results confirm that smart healthcare dimensions have distinct focuses, one on processes/technical and the other on people/social, configuring a sociotechnical system. More importantly, our results add more details, showing that these dimensions should be present and can integrate into an H4.0 system. These findings elucidate that H4.0 is not only about the introduction of digital technologies applied to processes, as is typically presented in the literature (e.g., Tortorella et al., 2022b; Raimo et al., 2023) but also should be evaluated from the perspective of people (Grover et al., 2018; Roscoe et al., 2019). Otherwise, hospital environments may face challenges and struggle to succeed with the adoption of digital technologies, as highlighted by Marcon et al. (2022).

The results indicate that for both dimensions of H4.0 to work within a hospital organizational environment, the entire organizational structure is necessary to support them. The H4.0 literature has sometimes presented a list of digital technologies that impact operations (Tortorella et al., 2022a; Tortorella et al., 2022b). However, implementation requires organizational support for digital technologies to succeed, especially for the healthcare smart working dimension, which is directly related to people (Kleiner, 2008; Baptista et al., 2020; Klein et al., 2024). Our results show that both healthcare smart operations and healthcare smart working need alignment with the digital strategic structure for implementation (thus supporting H1). Hospitals that structure their digital strategy have higher levels of H4.0 development. The digital strategy role in H4.0 implementation highlights that IT strategy and

business strategy for digital transformation are essential to conduct the process of consolidating a sociotechnical system (Chanias et al., 2019; Fischer et al., 2020). Furthermore, our findings show that digital strategy is an antecedent of the definition of digital resources, as hypothesized in H2. Prior research considered digital strategy and resources as a bundle of the strategic initiatives of digital transformation (Fischer et al., 2020; Verhoef et al., 2021), but our findings point to a sequential relationship of strategic alignment in which digital strategy is defined and will support the definition of digital resources. Also, we show that both internal and external digital resources matter for the development of this strategic alignment. This reinforces the qualitative prior findings from Fernandez-Vidal et al. (2022) that highlighted the need for governance structures able to work with internal and external sources that can support the digital transformation due to its complexity and variety of requirements that need to be attended.

We examined whether hospitals' level of network embeddedness within the healthcare ecosystem moderates the alignment between digital strategy and digital resources. The results reveal a significant positive moderating effect for external digital resources (supporting Hypothesis H3b), but not for internal resources (contrary to Hypothesis H3a). This distinction suggests that healthcare digital strategies aimed at leveraging external resources—particularly to support people-oriented digital transformation (smart working)—require strong ties to external actors. Although the digital transformation literature has emphasized the importance of external resources for digitalization (Franco et al., 2024), our findings indicate that, in the hospital context, this alignment is only effective when network embeddedness is high. That is, the ability to access and mobilize external resources depends not only on strategic intent but also on the hospital's level of connectivity within the broader healthcare network—especially in a sector where digital solutions are highly customized (Bavafa et al., 2021). In contrast, our results show no moderating effect of network embeddedness on the alignment between strategy and internal digital resources. While we initially expected that proximity to external actors would support internal digitalization

through knowledge diffusion and learning, the findings suggest otherwise. Internal resources appear to be more directly shaped by the hospital's internal strategic direction, without strong dependence on external connections. Given the complexity of healthcare systems (Stevens and Schaik, 2020) and the context-specific nature of internal capabilities (Mukherjee and Sinha, 2020), these resources may be inherently less susceptible to external influence. As a practical implication, hospitals aiming to develop smart working infrastructures should consider their position within healthcare networks. Those with high embeddedness can pursue strategies that leverage external digital resources. However, hospitals with low embeddedness may benefit more from focusing on internally driven digital resource development, thereby reducing uncertainties linked to limited external support. It should be emphasized that, in our analysis, we chose not to examine the moderating effect of network embeddedness on the relationship between digital strategy and the H4.0 smart dimensions because we understand that network embeddedness acts as a factor that enhances the relationship between strategy and digital resources and that, when integrated, can facilitate the implementation of the H4.0 dimensions. Our focus was to analyze its conditional effect, considering that this moderating effect does not act solely on digital strategy in isolation, but rather relationally, shaping how strategy and digital resources combine to generate digital capabilities that enable the hospital to develop H4.0.

We also tested whether the alignment between healthcare digital strategy and healthcare digital resources indeed allows for the development of higher levels of healthcare smart operations and healthcare smart working, as hypothesized in H3. We identified that healthcare digital resources (internal or external) mediate the relationship between healthcare digital strategy and the healthcare smart working dimension (thus supporting H3a), but we did not find statistical evidence for this mediation between digital strategy and smart operations. One possible explanation for this is grounded in the sociotechnical view (Clegg, 2000). Mostly, resources are more focused on supporting workers than on processes, regardless of how these resources are managed, internally or externally. Digital resources should be orchestrated to provide the best possible conditions so that technologies can be well-employed by workers (Piccoli et al., 2024). These resources create the condition for workers to be capable of dealing with new technologies, developing new skills, and integrating better their empirical knowledge of healthcare operations with the technology usage (Fischer et al., 2020; Gillani et al., 2024). On the other hand, healthcare smart operations involve only the introduction of technologies to perform tasks, most of the time without the presence of workers, making the need for resources to support them less critical. Of course, digital resources would be relevant for technology acquisition and implementation (Liu et al., 2011; Yeow et al., 2018; Verhoef et al., 2021), but not for its adoption which is dependent on the daily use in healthcare activities. Moreover, another possible explanation for the absence of evidence regarding this mediation may be that internal or external resources are less impactful due to regulatory, organizational, or technical constraints related to DT focused on process automation, which tends to be more flexible. In this case, certain types of digital resources to support technologies are not necessarily decisive, since process automation does not always involve human aspects, which might primarily require compliance with specific regulatory constraints, such as user data protection, or organizational and technical requirements, such as training and capacity-building for technology handling. Therefore, our findings suggest that the adoption of technologies aimed at healthcare smart operations may not necessarily demand a large concentration of institutional resources and efforts, as regulatory restrictions or internal organizational changes may not be strong enough to prevent, hinder, or condition the implementation of technological projects.

Our results also highlight a critical aspect, often overlooked but essential in digital strategies for the implementation of technologies such as IoT, electronic health records, and artificial intelligence (healthcare smart operations): security risks and cyber threats (Schmitt,

2023; Marcon et al., 2025). In a digital, intelligent, and interconnected environment, the act of automating processes requires the protection of sensitive data in the face of increasing threats (Schmitt, 2023). This concern is even more relevant in the healthcare sector, which is characterized by complex systems and non-standardized processes (Tortorella et al., 2020b; Stevens and Schaik, 2020; Uysal, 2022). Although our study does not delve deeply into this topic, the results reinforce the need to integrate cybersecurity structures — enabled by AI, for example, as proposed by Schmitt (2023) into digital strategies, to mitigate risks associated with the large volume of sensitive data (Aceto et al., 2020; Uysal, 2022) generated by both healthcare smart operations and healthcare smart working. Small vulnerabilities can lead to significant losses and compromise the confidentiality, integrity, and availability of organizational information, which requires robust strategic cybersecurity measures to protect both the organization, and the actors involved (Marcon et al., 2025). Therefore, there is a need to explore and develop cybersecurity resilience in the digital strategy dimension, a vital element for the security of healthcare organizations.

Finally, our findings align with the study by Edelman et al. (2005), which identified a positive association between the alignment of strategy and resources for performance improvement in organizations. They also resonate with the studies by Verhoef et al. (2021), presenting digital resources as a crucial strategic imperative for digital transformation. Furthermore, they are also related to the findings of a special edition presented by Baptista et al. (2020) on the strategic implications across various layers of technologies in the workplace and organizational structure. Gillani et al. (2024) also highlight the need for organizations to strategically navigate, including the establishment of an organizational structure as a facilitator for digital transformation. However, our results add a counterpoint to these studies by differentiating the role of resources in the strategic alignment, being different when it is associated with smart working or with smart operations.

6.1. Theoretical implications

Our study offers several contributions to existing literature and has various theoretical implications, as detailed below.

First, our results provide insights into organizational structures that may be associated with the implementation of digital transformation. In this regard, we extend previous literature that suggested considering strategic and management aspects in the adoption of digital technologies, both for processes and people. We empirically demonstrate these relationships in the context of healthcare. Additionally, existing literature has primarily focused on the implementation of digital technologies or how digital transformation is associated with improvements in operational performance. We broaden this perspective by highlighting the implementation of healthcare smart working and healthcare smart operations as outcomes of sociotechnical initiatives. While some researchers have found positive relationships between organizational factors and performance, we can further expand the sociotechnical view to achieve higher levels of digital technology implementation by considering social elements.

Second, we offer a theoretical contribution by combining the alignment perspective of digital strategy with digital resource management. Digital strategy adopts an “integrated view” approach, where both information technology strategy and business strategy are considered. Therefore, digital strategies in healthcare settings must be identified, understood, and translated into social initiatives that can support the implementation of H4.0 dimensions. We contribute to the theory by demonstrating that digital resources can serve as an operational means between digital strategy and H4.0 implementation. In sum, an organizational structure that aligns between digital strategy and digital resources facilitates the implementation of digital technologies to support healthcare workers, enabling hospitals to adequately consider the sociotechnical aspects of digital transformation in healthcare. Thus, we expand existing research by highlighting that alignment between

strategy and digital resources leads to higher levels of technological support for healthcare workers, adding the lens of the sociotechnical view.

Third, our study clarifies the role of organizational support in the digitization of healthcare environments. In this case, we demonstrate that this organizational support can emerge from the management of digital resources, enabling the provision of support for digital transformation. We show that even though digital resources may be managed in different ways, they can still be capable of providing this support for adoption and development, especially in the dimension of smart healthcare workers within H4.0. In other words, internally, hospitals can develop the necessary digital resources to support this process, just as hospitals can also outsource these resources since both the internal and external dimensions were positively associated with smart working. This represents possible pathways that reduce the pursuit of initiatives to support workers as an additional effort for their performance in carrying out activities. In this regard, an innovative finding is that organizational support through digital resources aligned with digital strategy can follow different management perspectives, considering not only the processes but also the people's perspective for changes in healthcare professionals' job roles. In sum, we encourage existing literature to place organizational support in the context of digital transformation, as well as its implications and changes in work in the digital era.

6.2. Managerial implications

Our study has several implications for managerial practice related to sociotechnical, organizational, and digitization aspects in healthcare, which are presented as follows.

Our findings show hospitals can consider not only the technocentric perspective to integrate and automate processes but also the sociotechnical view that involves people in implementing H4.0 in their operations. So far, the focus has been predominantly on processes. Expanding the potential of the H4.0 concept, sociotechnical initiatives should also be considered by hospitals to assist healthcare workers, both in the requirements for implementing technologies and changes in job roles and in maintaining the activities they perform over time, supported by digital technologies. To achieve this, hospitals must establish a well-defined digital strategy and have a set of digital resources to support the operationalization of these technologies by workers. When hospitals seek to increase interoperability and improve tasks performed by professionals in a high-complexity context, implementing H4.0 dimensions can help capture real-time information from both processes and patients to support decision-making, as well as worker support. Similarly, patients and society, among others, can be included in this context, allowing the integration of all stakeholders.

Secondly, organizational structure can play a significant role in mitigating potential obstacles to implementing H4.0. From a general perspective, our study recommends that healthcare managers structure the digital strategy to align with digital resources. This combination acts as an organizational pathway to implement digital technologies to support healthcare workers. Managers should also consider digital strategies that define operational targets for technology implementation, as digital strategy is also strongly related to digital technologies for healthcare smart operations. By considering these elements, managers can follow the path toward a sociotechnical organizational development of Healthcare 4.0.

Finally, our findings indicate that hospitals planning to implement smart working infrastructures should carefully assess their position within healthcare networks. Hospitals with strong external connections—such as partnerships with other institutions, technology providers, or public agencies—are better positioned to adopt strategies that rely on external digital resources. These networks can offer critical support and expertise during digital transformation efforts. In contrast, hospitals with limited embeddedness in such networks may benefit more from focusing on internally driven resource development. This approach

can reduce uncertainty and increase control over implementation, ensuring that smart working initiatives are grounded in the hospital's own capabilities.

6.3. Limitations and future research

This study presents some methodological limitations that point to opportunities for future research. Regarding the sample data of the study, all interviewees were from hospitals located in Brazil, a country classified as an emerging economy. This may limit the generalizability of our findings to different socioeconomic contexts. While we have identified significant and consistent relationships, we suggest that this study could be conducted in other socioeconomic contexts, involving hospitals from various locations. This would allow for an international comparative analysis that could contribute to a global understanding, aiding in identifying international differences and strategies for generating social value. Also, our constructs of digital resources evaluate the resources used by hospitals to support the digitization of their operational processes and workers' activities. However, we are limited to considering only which resources are appropriate for this support. As digital transformation progresses, other types of resources and/or workers' capabilities may become necessary to support the implementation and use of different new technologies. For example, the use of generative AI as a medical support tool is a promising field for future research. Therefore, identifying which new digital resources and capabilities are needed for this and other technologies can guide further future investigations.

Lastly, our constructs of digital strategy and digital resources allowed us to analyze the relationship between them in implementing the H4.0 smart dimensions. However, other sociotechnical aspects, such as worker learning analysis, technostress evaluations, and the development of new digital skills and capabilities, for example, could also be considered. This enables the development of new research. Additionally, future studies could also investigate the role of this alignment between digital strategy and digital resources in the performance of healthcare workers. Moreover, another limitation of our study concerns the scope of the questionnaire, which was restricted to organizational-level questions, without addressing individual-level aspects within the hospitals. For future research, it would be interesting to include additional questions specifically directed at healthcare professionals, exploring the complexity of their tasks, their digital experience, and information about their profiles. We also suggest grouping participants according to their respective hospital departments to deepen their understanding of the factors that influence the success of HC4.0.

CRedit authorship contribution statement

Lucas Miguel Alencar de Morais Correia: Writing – original draft, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Paulo Renato de Sousa:** Visualization, Conceptualization. **Alejandro G. Frank:** Writing – review & editing, Validation, Supervision, Project administration, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

All authors of this paper declare that they have no conflict of interest.

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Appendix A. Questionnaire structure

Questionnaire items to assess Healthcare Digital Strategy (STRATEGY)

(Adapted from Nasiri et al. (2020) and Verhoef et al. (2021)). Concordance Likert scale: 1 - strongly disagree to 5 - strongly agree. CFI = 0.990; TLI = 0.983; RMSEA = 0.070; AVE = 0.685; CR = 0.928; Cronbach = 0.928. Factor loadings are shown in parentheses.

- a) Our hospital aims to excel in the efficiency of its processes through the use of digital technologies (0.752).
- b) Our hospital seeks to collect and integrate data from different internal areas through the use of digital technologies (0.816).
- c) Our hospital aims to create a strong communication network among different internal areas through the use of digital technologies (0.865).
- d) Our hospital aims to provide more efficient medical services through the use of digital technologies (0.884).
- e) Our hospital seeks to enhance contact with patients through the use of digital technologies (0.845).
- f) Our hospital aims to improve its relationship with suppliers through the use of digital technologies (0.795).

Questionnaire items to assess Internal Digital Healthcare Resources (INTERNAL)

(Adapted from Barney, 1986, 1991; Franco et al., 2024). Concordance Likert scale: 1 - strongly disagree to 5 - strongly agree. CFI = 0.995; TLI = 0.989; RMSEA = 0.043; AVE = 0.506; CR = 0.835; Cronbach = 0.831. Factor loadings are shown in parentheses.

- a) Our hospital has the necessary knowledge to develop healthcare digital transformation (0.730).
- b) Our hospital has the necessary information to develop healthcare digital transformation (0.759).
- c) Our hospital has the necessary human resources to develop healthcare digital transformation (0.796).
- d) Our hospital has the necessary financial resources to develop healthcare digital transformation (0.641).
- e) Our hospital has the necessary organizational culture to develop its own healthcare digital transformation (0.614).

Questionnaire items to assess External Digital Healthcare Resources (EXTERNAL)

(Adapted from Dyer and Singh, 1998; Dyer et al., 2018; Franco et al., 2024). Concordance Likert scale: 1 - strongly disagree to 5 - strongly agree. CFI = 0.968; TLI = 0.947; RMSEA = 0.080; AVE = 0.444; CR = 0.826; Cronbach = 0.823. Factor loadings are shown in parentheses.

- a) Our hospital uses complementary knowledge from external partners to develop healthcare digital transformation (0.776).
- b) Our hospital has the necessary information from external partners to develop healthcare digital transformation (0.693).
- c) Our hospital utilizes human resources from external partners to develop healthcare digital transformation (0.641).
- d) Our hospital utilizes technological resources from external partners to develop digital transformation in healthcare (0.618).
- e) Our hospital utilizes financial resources from external partners to develop healthcare digital transformation (0.582).
- f) Our hospital has a culture of collaboration with external partners for the development of healthcare digital transformation (0.671).

Questionnaire items to assess Healthcare Network Embeddedness (EMBEDDEDNESS)

(Adapted from Xing et al., 2023; Chen et al., 2024; Chen et al., 2025). Concordance Likert scale: 1 - strongly disagree to 5 - strongly agree. CFI = 0.987; TLI = 0.979; RMSEA = 0.072; AVE = 0.637; CR = 0.913;

Cronbach = 0.913. Factor loadings are shown in parentheses.

- a) Our hospital is located in an environment close to healthcare suppliers that develop innovative solutions for the healthcare sector (0.792).
- b) Our hospital is located in an environment close to other hospitals and clinics that develop innovative solutions for the healthcare sector (0.826).
- c) Our hospital is located in an environment close to healthcare operators (insurers, health plans, agreements, etc.) that develop innovative solutions for the healthcare sector (0.757).
- d) Our hospital is located in an environment close to public actors that develop innovative solutions for the healthcare sector (0.766).
- e) Our hospital is located in an environment close to universities and research centers that develop innovative solutions for the healthcare sector (0.862).
- f) Our hospital is located in an environment where there are collaboration networks and mechanisms for the transfer of knowledge and technologies for innovation in the healthcare sector (0.782).

Questionnaire items to assess healthcare technologies for smart operations and smart working

(Adapted from Frank et al. (2019), Meindl et al. (2021), Dornelles et al. (2022), Tortorella et al. (2022a, 2022b) and Deloitte (2020)). Concordance Likert scale: 1 - never to 5 - very frequently.

- a) Our hospital uses big data/analytics to support activities.
- b) Our hospital uses artificial intelligence/machine learning to support activities.
- c) Our hospital uses blockchain to support activities.
- d) Our hospital uses augmented or virtual reality to support activities.
- e) Our hospital uses collaborative robotics (e.g., surgical robots, cobots, outpatient care, AGVs, AMRs, etc.) to support activities.
- f) Our hospital uses wireless biomedical/digital sensors to support activities.
- g) Our hospital uses digital patient relationship applications to support activities.
- h) Our hospital uses cloud computing in hospital processes.
- i) Our hospital uses mobile applications to support activities.
- j) Our hospital uses corporate management systems (e.g., ERP) for the integration of internal data and with suppliers.
- k) Our hospital uses electronic health records (e.g., electronic prescription, electronic medical records) to perform operational activities.
- l) Our hospital uses digital traceability (e.g., RFID, QR Code, etc.) to carry out activities.
- m) Our hospital uses simulation to support its activities.
- n) Our hospital uses telemedicine to provide services to patients.

Questionnaire items for control variables

- a) Number of beds in your hospital.
- b) What is the approximate number of employees working at your hospital.
- c) How many years ago was your hospital founded.
- d) Ownership of your hospital (two dummies):
Tier 1 – Teaching;
Tier 2 – Public;
Tier 3 – Private.
- e) Your role within your hospital:
Middle manager (e.g., supervisor or coordinator)
Senior manager (e.g., manager or director).

Data availability

The authors do not have permission to share data.

References

- Aceto, G., Persico, V., Pescapé, A., 2020. Industry 4.0 and health: internet of things, big data, and cloud computing for healthcare 4.0. *J. Ind. Inf. Integr.* 18 (100129), 100129. <https://doi.org/10.1016/j.jii.2020.100129>.
- Baiyere, A., Salmela, H., Tapanainen, T., 2020. Digital transformation and the new logics of business process management. *Eur. J. Inf. Syst.* 29 (3), 238–259. <https://doi.org/10.1080/0960085X.2020.1718007>.
- Baptista, J., Stein, M.K., Klein, S., Watson-Manheim, M.B., Lee, J., 2020. Digital work and organisational transformation: emergent digital/human work configurations in modern organisations. *J. Strateg. Inf. Syst.* 29 (2), 101618. <https://doi.org/10.1016/j.jsis.2020.101618>.
- Barney, J.B., 1986. Organizational culture: can it be a source of sustained competitive advantage? *Acad. Manag. Rev.* 11 (3), 656. <https://doi.org/10.2307/258317>.
- Barney, J., 1991. Firm resources and sustained competitive advantage. *J. Manag.* 17 (1), 99–120. <https://doi.org/10.1177/014920639101700108>.
- Basile, L.J., Carbonara, N., Panniello, U., Pellegrino, R., 2024. The role of big data analytics in improving the quality of healthcare services in the Italian context: the mediating role of risk management. *Technovation* 133, 103010. <https://doi.org/10.1016/j.technovation.2024.103010>.
- Bavafa, H., Savin, S., Terwiesch, C., 2021. Customizing primary care delivery using E-visits. *Prod. Oper. Manag.* 30 (11), 4306–4327. <https://doi.org/10.1111/poms.13528>.
- Beane, M., 2019. Shadow learning: building robotic surgical skill when approved means fail. *Adm. Sci. Q.* 64 (1), 87–123. <https://doi.org/10.1177/0001839217751692>.
- Cagliano, R., Canterino, F., Longoni, A., Bartezzaghi, E., 2019. The interplay between smart manufacturing technologies and work organization: the role of technological complexity. *Int. J. Oper. Prod. Manag.* 39 (6/7/8), 913–934. <https://doi.org/10.1108/IJOPM-01-2019-0093>.
- Cannavacciuolo, L., Capaldo, G., Ponsiglione, C., 2023. Digital innovation and organizational changes in the healthcare sector: multiple case studies of telemedicine project implementation. *Technovation* 120, 102550. <https://doi.org/10.1016/j.technovation.2022.102550>.
- Cerchione, R., Centobelli, P., Riccio, E., Abbate, S., Orpallo, E., 2023. Blockchain's coming to hospital to digitalize healthcare services: designing a distributed electronic health record ecosystem. *Technovation* 120, 102480, 2023.
- Chaniyas, S., Myers, M.D., Hess, T., 2019. Digital transformation strategy making in pre-digital organizations: the case of a financial services provider. *J. Strateg. Inf. Syst.* 28 (1), 17–33. <https://doi.org/10.1016/j.jsis.2018.11.003>.
- Chaudhuri, A., Nasereldin, H., Narayanamurthy, G., 2023. Healthcare 3D printing service innovation: resources and capabilities for value co-creation. *Technovation* 121, 102596. <https://doi.org/10.1016/j.technovation.2022.102596>.
- Chen, Yihua, Yang, S., Zhang, Z., Shen, H., 2024. How does digital enablement affect product customization? The roles of innovation capability and network embeddedness. *Technol. Forecast. Soc. Change* 201 (123272), 123272. <https://doi.org/10.1016/j.techfore.2024.123272>.
- Chen, Y., Li, B., Huo, B., 2025. Building operational resilience through digitalization: the roles of supply chain network position. *Technol. Forecast. Soc. Change* 211 (123918), 123918. <https://doi.org/10.1016/j.techfore.2024.123918>.
- Clegg, C.W., 2000. Sociotechnical principles for system design. *Appl. Ergon.* 31, 463–477. [https://doi.org/10.1016/S0003-6870\(00\)00009-0](https://doi.org/10.1016/S0003-6870(00)00009-0).
- Cohen, J., Cohen, P., West, S.G., Aiken, L.S., 2003. *Applied Multiple Regression/Correlation Analysis for the Behavioral Sciences*, third ed. Routledge.
- Constantiou, I.D., Kallinikos, J., 2015. New games, new rules: big data and the changing context of strategy. *J. Inf. Technol.* 30 (1), 44–57. <https://doi.org/10.1057/jit.2014.17>.
- Correia, L.M.A. de M., Frank, A.G., 2025. Structuring resources in healthcare digital transformation: a comparison across public, private and research hospitals. *Technovation* 147 (103320), 103320. <https://doi.org/10.1016/j.technovation.2025.103320>.
- Dahlke, J., Beck, M., Kinne, J., Lenz, D., Dehghan, R., Wörter, M., Ebersberger, B., 2024. Epidemic effects in the diffusion of emerging digital technologies: evidence from artificial intelligence adoption. *Res. Policy* 53 (2), 104917. <https://doi.org/10.1016/j.respol.2023.104917>.
- Dalenogare, L.S., Le Dain, M.-A., Ayala, N.F., Pezzotta, G., Frank, A.G., 2023. Building digital servitization ecosystems: an analysis of inter-firm collaboration types and social exchange mechanisms among actors. *Technovation* 124, 102756. <https://doi.org/10.1016/j.technovation.2023.102756>.
- Deloitte, 2020. Digital Transformation: Shaping the Future of European Healthcare. Retrieved from: <https://www2.deloitte.com/content/dam/Deloitte/uk/Documents/life-sciences-health-care/deloitte-uk-shaping-the-future-of-european-healthcare.pdf>.
- Denicolai, S., Previtali, P., 2023. Innovation strategy and digital transformation execution in healthcare: the role of the general manager. *Technovation* 121, 102555. <https://doi.org/10.1016/j.technovation.2022.102555>.
- Dicuonzo, G., Donofrio, F., Fusco, A., Shini, M., 2023. Healthcare system: moving forward with artificial intelligence. *Technovation* 120, 102510. <https://doi.org/10.1016/j.technovation.2022.102510>.
- Dornelles, J. de Assis, Ayala, N.F., Frank, A.G., 2022. Smart working in industry 4.0: how digital technologies enhance manufacturing workers' activities. *Comput. Ind. Eng.* 163, 107804. <https://doi.org/10.1016/j.cie.2021.107804>.
- Dornelles, J. de Assis, Ayala, N.F., Frank, A.G., 2023. Collaborative or substitutive robots? Effects on workers' skills in manufacturing activities. *Int. J. Prod. Res.* 61 (22), 7922–7955. <https://doi.org/10.1080/00207543.2023.2240912>.
- Dyer, J.H., Singh, H., 1998. The relational view: cooperative strategy and sources of interorganizational competitive advantage. *Acad. Manag. Rev.* 23 (4), 660–679. <https://doi.org/10.2307/259056>.
- Dyer, J.H., Singh, H., Hesterly, W.S., 2018. The relational view revisited: a dynamic perspective on value creation and value capture. *Strateg. Manag. J.* 39 (12), 3140–3162. <https://doi.org/10.1002/smj.2785>.
- Edelman, L.F., Brush, C.G., Manolova, T., 2005. Co-alignment in the resource–performance relationship: strategy as mediator. *J. Bus. Ventur.* 20 (3), 359–383. <https://doi.org/10.1016/j.jbusvent.2004.01.004>.
- Enrique, D.V., Marodin, G.A., Santos, F.B.C., Frank, A.G., 2023. Implementing industry 4.0 for flexibility, quality, and productivity improvement: technology arrangements for different purposes. *Int. J. Prod. Res.* 61 (20), 7001–7026. <https://doi.org/10.1080/00207543.2022.2142689>.
- Fernandez-Vidal, J., Perotti, A.F., Gonzalez, R., Gasco, J., 2022. Managing digital transformation: the view from the top. *J. Bus. Res.* 152, 29–41. <https://doi.org/10.1016/j.jbusres.2022.07.020>.
- Fischer, M., Imgrund, F., Janiesch, C., Winkelman, A., 2020. Strategy archetypes for digital transformation: defining meta objectives using business process management. *Inf. Manag.* 57 (5), 103262. <https://doi.org/10.1016/j.im.2019.103262>.
- Fornell, C., Larcker, D.F., 1981. Evaluating structural equation models with unobservable variables and measurement error. *J. Market. Res.* 18 (1), 39. <https://doi.org/10.1177/00224378101800104>.
- Franco, W.C., Benitez, G.B., Sousa, P.R., Neto, F.J.K., Frank, A.G., 2023. A contingency-configurational view of purchasing operations: the mediating role between supplier relationship and firm performance. *J. Purch. Supply Manag.* 29, 100815. <https://doi.org/10.1016/j.pursup.2022.100815>.
- Franco, C.W., Benitez, G.B., de Sousa, P.R., Kliemann Neto, F.J., Frank, A.G., 2024. Managing resources for digital transformation in supply chain integration: the role of hybrid governance structures. *Int. J. Prod. Econ.* 278 (109428), 109428. <https://doi.org/10.1016/j.ijpe.2024.109428>.
- Frank, A.G., Dalenogare, L.S., Ayala, N.F., 2019. Industry 4.0 technologies: implementation patterns in manufacturing companies. *Int. J. Prod. Econ.* 210, 15–26. <https://doi.org/10.1016/j.ijpe.2019.01.004>.
- Gillani, F., Chatha, K.A., Jajja, S.S., Cao, D., Ma, X., 2024. Unpacking digital transformation: identifying key enablers, transition stages and digital archetypes. *Technol. Forecast. Soc. Chang.* 203, 123335. <https://doi.org/10.1016/j.techfore.2024.123335>.
- Gnyawali, D.R., Madhavan, R., 2001. Cooperative networks and competitive dynamics: a structural embeddedness perspective. *Acad. Manag. Rev.* 26 (3), 431–445.
- Granovetter, M., 1985. Economic action and social structure: the problem of embeddedness. *Am. J. Sociol.* 91 (3), 481–510. <http://www.jstor.org/stable/2780199>.
- Grover, P., Kar, A.K., Davies, G., 2018. “Technology enabled health” – insights from twitter analytics with a socio-technical perspective. *Int. J. Inf. Manag.* 43, 85–97. <https://doi.org/10.1016/j.ijinfomgt.2018.07.003>.
- Hair, J.F., Bask, W.C., Babin, B.J., Anderson, R.E., 2018. *Multivariate Data Analysis*, 8th ed. Cengage.
- Hayes, A.F., 2017. *Introduction to Mediation, Moderation, and Conditional Process Analysis. A Regression-based Approach*. Guilford Press.
- Kearns, G., Lederer, A., 2000. The effect of strategic alignment on the use of IS-based resources for competitive advantage. *J. Strateg. Inf. Syst.* 9 (4), 265–293. [https://doi.org/10.1016/S0963-8687\(00\)00049-4](https://doi.org/10.1016/S0963-8687(00)00049-4).
- Ketokivi, M., 2019. Avoiding bias and fallacy in survey research: a behavioral multilevel approach. *J. Oper. Manag.* 65 (4), 380–402. <https://doi.org/10.1002/joom.1011>.
- Klein, S., Watson-Manheim, M.B., 2021. The (re-)configuration of digital work in the wake of profound technological innovation: constellations and hidden work. *Inf. Organ.* 31 (4), 100377. <https://doi.org/10.1016/j.infoandorg.2021.100377>.
- Klein, S.P., Spieth, P., Söllner, M., 2024. Employee acceptance of digital transformation strategies: a paradox perspective. *J. Prod. Innov. Manag.* 2024, 1–23. <https://doi.org/10.1111/jpim.12722>.
- Kleiner, B.M., 2008. Macroergonomics: work system analysis and design. *Hum. Factors* 50 (3), 461–467. <https://doi.org/10.1518/001872008X288501>.
- Klinker, K., Wiesche, M., Krcmar, H., 2020. Digital transformation in health care: augmented reality for hands-free service innovation. *Inf. Syst. Front.* 22, 1419–1431. <https://doi.org/10.1007/s10796-019-09937-7>.
- Kraus, S., Schiavone, F., Pluzhnikova, A., Invernizzi, A.C., 2021. Digital transformation in healthcare: analyzing the current state-of-research. *J. Bus. Res.* 123, 557–567. <https://doi.org/10.1016/j.jbusres.2020.10.030>.
- Lerman, L.V., Benitez, G.B., Müller, J.M., de Sousa, P.R., Frank, A.G., 2022. Smart green supply chain management: a configurational approach to enhance green performance through digital transformation. *Supply Chain Manag. Int. J.* 27 (7), 147–176. <https://doi.org/10.1108/SCM-02-2022-0059>.
- Liao, Y., Deschamps, F., Loures, E.F.R., Ramos, L.F.P., 2017. Past, present and future of industry 4.0: a systematic literature review and research agenda proposal. *Int. J. Prod. Res.* 55 (12), 3609–3629. <https://doi.org/10.1080/00207543.2017.1308576>.
- Lin, N., 2001. *Social Capital: A Theory of Social Structure and Action*. Cambridge University Press, New York, NY.
- Liu, D., Chen, S., Chou, T., 2011. Resource fit in digital transformation: lessons learned from the CBC bank global e-banking project. *Manag. Decis.* 49 (10), 1728–1742. <https://doi.org/10.1108/00251741111183852>.
- Lyu, Y., Zhu, Y., Han, S., He, B., Bao, L., 2020. Open innovation and innovation “radicalness”—the moderating effect of network embeddedness. *Technol. Soc.* 62 (101292), 101292. <https://doi.org/10.1016/j.techsoc.2020.101292>.

- Marcon, E., Soliman, M., Gerstlberger, W., Frank, A.G., 2022. Sociotechnical factors and Industry 4.0: an integrative perspective for the adoption of smart manufacturing technologies. *J. Manuf. Technol. Manag.* 33 (2), 259–286. <https://doi.org/10.1108/JMTM-01-2021-0017>.
- Marcon, E., Marodin, G.A., Frank, A.G., 2024. Combining organizational and social factors to support industry 4.0 implementation: a sociotechnical and configurational analysis of technology adopters. *IEEE Trans. Eng. Manag.* 72, 146–160. <https://doi.org/10.1109/tem.2024.3506991>.
- Marcon, E., Soliman, M., Ayala, N.F., Sturgeon, T., Frank, A.G., 2025. A system view of how balancing sociotechnical factors in industry 4.0 implementation: evidence from interviews and ethnographic research. *Comput. Ind. Eng.* 203 (111047), 111047. <https://doi.org/10.1016/j.cie.2025.111047>.
- Mazzola, E., Perrone, G., Kamuriwo, D.S., 2015. Network embeddedness and new product development in the biopharmaceutical industry: the moderating role of open innovation flow. *Int. J. Prod. Econ.* 160, 106–119. <https://doi.org/10.1016/j.ijpe.2014.10.002>.
- Meindl, B., Ayala, N.F., Mendonça, J., Frank, A.G., 2021. The four smarts of Industry 4.0: Evolution of ten years of research and future perspectives. *Technol. Forecast. Soc. Change* 168, 120784.
- Miroshnychenko, I., Strobl, A., Matzler, K., De Massis, A., 2021. Absorptive capacity, strategic flexibility, and business model innovation: empirical evidence from Italian SMEs. *J. Bus. Res.* 130, 670–682. <https://doi.org/10.1016/j.jbusres.2020.02.015>.
- Mukherjee, U.K., Sinha, K.K., 2020. Robot-assisted surgical care delivery at a hospital: policies for maximizing clinical outcome benefits and minimizing costs. *J. Oper. Manag.* 66 (12), 227–256. <https://doi.org/10.1002/joom.1058>.
- Nasiri, M., Ukko, J., Saunila, M., Rantala, T., 2020. Managing the digital supply chain: the role of smart technologies. *Technovation* 96–97, 102121. <https://doi.org/10.1016/j.technovation.2020.102121>.
- Naslund, J.A., Shidhaye, R., Patel, V., 2019. Digital technology for building capacity of nonspecialist health workers for Task sharing and scaling up mental health care globally. *Harv. Rev. Psychiatry* 27 (3), 181–192. <https://doi.org/10.1097/HRP.0000000000000217>.
- Oliveira, M., Zancul, E., Salerno, M.S., 2024. Capability building for digital transformation through design thinking. *Technol. Forecast. Soc. Chang.* 198, 122947. <https://doi.org/10.1016/j.techfore.2023.122947>.
- Parmigiani, A., 2007. Why do firms both make and buy? An investigation of concurrent sourcing. *Strateg. Manag. J.* 28 (3), 285–311. <https://doi.org/10.1002/smj.580>.
- Piccoli, G., Grover, V., Rodriguez, J., 2024. Digital transformation requires digital resource primacy: clarification and future research directions. *J. Strateg. Inf. Syst.* 33 (2), 101835. <https://doi.org/10.1016/j.jsis.2024.101835>.
- Podsakoff, P.M., MacKenzie, S.B., Lee, J.Y., Podsakoff, N.P., 2003. Common method biases in behavioral research: a critical review of the literature and recommended remedies. *J. Appl. Psychol.* 88 (5), 879–903. <https://psycnet.apa.org/doi/10.1037/0021-9010.88.5.879>.
- Preacher, K.J., Hayes, A.F., 2008. Asymptotic and resampling strategies for assessing and comparing indirect effects in multiple mediator models. *Behav. Res. Methods* 40 (3), 879–891. <https://doi.org/10.3758/BRM.40.3.879>.
- Raimo, N., Turi, I., Albero, F., Vitolla, F., 2023. The drivers of the digital transformation in the healthcare industry: an empirical analysis in Italian hospitals. *Technovation* 121, 102558. <https://doi.org/10.1016/j.technovation.2022.102558>.
- Roscoe, S., Cousins, P.D., Handfield, R., 2019. The microfoundations of an operational capability in digital manufacturing. *J. Oper. Manag.* 65 (8), 774–793. <https://doi.org/10.1002/joom.1044>.
- Schmitt, M., 2023. Securing the digital world: protecting smart infrastructures and digital industries with artificial intelligence (AI)-enabled malware and intrusion detection. *J. Ind. Inf. Integr.* 36 (100520), 100520. <https://doi.org/10.1016/j.jii.2023.100520>.
- Stevens, M., Schaik, J., 2020. Implementing new technologies for complex care: the role of embeddedness factors in team learning. *J. Oper. Manag.* 66 (1–2), 112–134. <https://doi.org/10.1002/joom.1034>.
- Taherdoost, H., August 10, 2016. Validity and reliability of the research instrument; how to test the validation of a questionnaire/survey in a research. How to test the validation of a questionnaire/survey in a research. Available at SSRN <https://ssrn.com/abstract=3205040>.
- Thuemmler, C., Bai, C., 2017. Health 4.0: Application of industry 4.0 design principles in future asthma management. In: *Healthcare 4.0: How Virtualization and Big Data are Revolutionizing Healthcare*. Springer, Cham, London, pp. 23–37. https://doi.org/10.1007/978-3-319-47617-9_2.
- Tortorella, G.L., Fogliatto, F.S., Mac Cawley Vergara, A., Vassolo, R., Sawhney, R., 2020a. Healthcare 4.0: trends, challenges and research directions. *Prod. Plan. Control* 31 (15), 1245–1260. <https://doi.org/10.1080/09537287.2019.1702226>.
- Tortorella, G.L., Fogliatto, F.S., Esposto, K.F., Vergara, A.M.C., Vassolo, R., Mendoza, D. T., Narayanamurthy, G., 2020b. Effects of contingencies on healthcare 4.0 technologies adoption and barriers in emerging economies. *Technol. Forecast. Soc. Chang.* 156, 120048. <https://doi.org/10.1016/j.techfore.2020.120048>.
- Tortorella, G.L., Fogliatto, F.S., Esposto, K.F., Vergara, A.M.C., Vassolo, R., Mendoza, D. T., Narayanamurthy, G., 2022a. Measuring the effect of healthcare 4.0 implementation on hospitals' performance. *Prod. Plan. Control* 33 (4), 386–401. <https://doi.org/10.1080/09537287.2020.1824283>.
- Tortorella, G.L., Fogliatto, F.S., Kurnia, S., Thürer, M., Capurro, D., 2022b. Healthcare 4.0 digital applications: an empirical study on measures, bundles and patient-centered performance. *Technol. Forecast. Soc. Chang.* 181, 121780. <https://doi.org/10.1016/j.techfore.2022.121780>.
- Tortorella, G.L., Fogliatto, F.S., Saurin, T.A., Tonetto, L.M., McFarlane, D., 2022c. Contributions of healthcare 4.0 digital applications to the resilience of healthcare organizations during the COVID-19 outbreak. *Technovation* 111, 102379. <https://doi.org/10.1016/j.technovation.2021.102379>.
- Uysal, M.P., 2022. Machine learning-enabled healthcare information systems in view of industrial information integration engineering. *J. Ind. Inf. Integr.* 30 (100382), 100382. <https://doi.org/10.1016/j.jii.2022.100382>.
- Verhoef, P.C., Broekhuizen, T., Bart, Y., Bhattacharya, A., Dong, J.Q., Fabian, N., Haenlein, M., 2021. Digital transformation: a multidisciplinary reflection and research agenda. *J. Bus. Res.* 122, 889–901. <https://doi.org/10.1016/j.jbusres.2019.09.022>.
- Vial, G., 2019. Understanding digital transformation: a review and a research agenda. *J. Strateg. Inf. Syst.* 28 (2), 118–144. <https://doi.org/10.1016/j.jsis.2019.01.003>.
- Wang, Y., Kung, L., Wang, W., Cegielski, C., 2018. An integrated big data analytics-enabled transformation model: application to health care. *Inf. Manag.* 55 (1), 64–79. <https://doi.org/10.1016/j.im.2017.04.001>.
- Xing, X., Chen, T., Yang, X., Liu, T., 2023. Digital transformation and innovation performance of China's manufacturers? A configurational approach. *Technol. Soc.* 75 (102356), 102356. <https://doi.org/10.1016/j.techsoc.2023.102356>.
- Yeow, A., Soh, C., Hansen, R., 2018. Aligning with new digital strategy: a dynamic capabilities approach. *J. Strateg. Inf. Syst.* 27 (1), 43–58. <https://doi.org/10.1016/j.jsis.2017.09.001>.
- Zhao, X., Lynch, J.G., Chen, Q., 2010. Reconsidering baron and kenny: myths and truths about mediation analysis. *J. Consum. Res.* 37 (2), 197–206. <https://doi.org/10.1086/651257>.
- Zimmer, M.P., Baiyere, A., Salmela, H., 2023. Digital workplace transformation: subtraction logic as deinstitutionalising the taken-for-granted. *J. Strateg. Inf. Syst.* 32 (1), 101757. <https://doi.org/10.1016/j.jsis.2023.101757>.

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